UPDATED
SPRING 2025

Greenwich University

COURSE OUTLINE

Course Title:	**Applied Physics**
Course Code:	**GCAP 314**
Credit Hours:	2
Department:	Computer Science
Faculty:	Management Sciences & Information Studies

Course Description

The course introduces students with the basic concept of Physics and electronics. Students are also taught Physics laws and other associate topics to prepare them for the advanced level courses in this area. The focus of the course on electric force and its applications and related problems, conservation of charge, charge quantization, Electric fields due to point charge and lines of force and many other useful topics.

Program Learning Outcomes (PLOs)

- Graduates will demonstrate a solid understanding of core computer science concepts, including algorithms, data structures, software engineering, databases, and operating systems, and apply these principles in solving complex computational problems.
- Graduates will possess the ability to analyze, design, and implement efficient solutions to computational problems by applying mathematical and algorithmic principles, and using appropriate computing techniques and tools.
- Graduates will be proficient in designing, developing, testing, and maintaining software systems, using current programming languages, frameworks, and development methodologies while adhering to industry best practices.
- Graduates will effectively communicate technical information, both orally and in writing, and collaborate productively as members or leaders of diverse, multidisciplinary teams in professional settings.
- Graduates will recognize the ethical, legal, and societal implications of computing technologies, and demonstrate responsible conduct by adhering to professional standards and promoting sustainable and inclusive solutions in the field of computer science.

Course Objectives

The course aims to provide students with a comprehensive understanding of electromagnetic theory and optics, focusing on electric forces, fields, and potentials, as well as the principles of magnetism and wave behavior. Students will learn to apply these concepts to solve real-world problems in physics and engineering.

Course Learning Outcomes (CLOs)

- Identify and describe fundamental concepts of electric forces, fields, and potentials
- Explain the principles of Gauss' Law and its applications to various charge distributions
- Evaluate the behavior of light through reflection, refraction and diffraction, using wave theory to address practical optical phenomena.

CLOs PLO Mapping

CLOs	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5
1	X				
2	X	X			
3	X		X		

Course Schedule

Week	Topic	Corresponding CLOs	Teaching Methodology	Assessment	Teaching Material	SDGs
1	Electric Force and Charge	1	Lecture, Problem-Solving	Quiz, Assignments	Textbook, Simulations	4
2	Electric Fields Due to Point Charges and Lines of Force	1	Lecture, Demonstrations	Problem Sets	Interactive Models	4, 9
3	Dipole in an Electric Field and Flux of Electric Field	1	Concept Mapping, Problem-Solving	Homework, Group Discussion	Research Papers	4
4	Gauss' Law and Its Applications	2	Lecture, Whiteboard Exercises	Quiz, MCQs	Textbook, Simulations	4
5	Electric Potential Energy and Electric Potentials	2	Lecture, Peer Discussion	Quiz	Lecture Notes, Videos	4, 7
6	Potential Due to Point and Continuous Charge Distribution	2	Problem-Solving, Peer Teaching	Group Work, Assignment	Case Studies, Simulations	4, 9
7	Calculating the Field from Potential and Electric Current	2	Lab Demonstration, Whiteboard Problem-Solving	Lab Report, Test	Lab Equipment, Simulations	4, 9
	Mid-Term Exams	-	-	-	-	-

8	Resistance, Resistivity, Conductivity, and Ohm's Law	CLO 1, CLO 3	Lecture, Hands-on Lab	Quiz, Problem Sets	Circuit Simulations	4, 7, 9
9	The Hall Effect and Magnetic Force on Current	CLO 1, CLO 3	Lecture, Case Studies	MCQs, Assignments	Research Papers	4, 9
10	Ampere's Law, Solenoid, and Toroids	CLO 1	Interactive Simulations	Quiz, Problem Sets	Virtual Labs, Textbook	4, 9
11	Faraday's Law of Induction and Lenz's Law	CLO 1, CLO 3	Lecture, Experiments	Quiz, Lab Report	Electromagnetic Experiments	4, 9
12	Electromagnetism Equations and Induced Magnetic Fields	CLO 1, CLO 3	Lecture, Case Study Discussion	Homework, Group Work	Research Articles	4, 9
13	Reflection, Refraction, and Interference	CLO 3	Lab Demonstrations, Simulations	Lab Report, Quiz	Optical Experiment Kits	4, 9
14	Diffraction, Polarization, and Related Problems	CLO 3	Lecture, Hands-on Experiment	Quiz, Presentation	Simulation Software, Papers	4, 9
	Final Exams	-	-	-	-	-

Reading Material

Textbooks:

- Fundamentals of Physics (Extended), 10th edition, Resnick and Walker

Reference Books:

- Narciso Garcia, Arthur Damask, Steven Schwarz., "Physics for Computer Science Students", Springer Verlag, 1998.

Assessment & Marking Scheme

Assessment Type	**Weightage**
Quizzes	10%
Assignments (Practical Work)	10%
Final Presentation/ Class Participation	10%
Mid-Term Exams	30%
Final Exam	40%

Guideline for Assessment & Marking Scheme

Quizzes	**10 Marks**

- A minimum of 5 quizzes during the whole semester are required
- The format of the quiz must be at least 10 MCQs or 01 Long Question based on the previous classes held.

Assignments	**10 Marks**

The assessment criteria for assignments and the final project will be based on the following table. This table evaluates key areas such as practical application, research and analysis, presentation skills, innovation, and the ability to reflect on and apply learning. Portfolio marks, allocated on the white card, will be determined by the students' performance across these areas, reflecting their overall engagement and progress throughout the course. The assessment will ensure that students are graded fairly based on their ability to demonstrate understanding and apply ethical concepts effectively.

Category	Objective	Preparation	Execution	Assessment

Final Presentation/ Project		**10 Marks**
Criteria	**Details**	**Marks**
Title Slide	- Presentation title - Student Name - Course Name - Instructor's Name - Date of Presentation	1
Introduction	- Greeting and self-introduction - Overview of topic and its relevance - Purpose and objectives of the presentation	1-2
Main Content	- Context/Background: Brief but comprehensive overview - Key Concepts/Findings: Present theories/data with visuals - Analysis/Discussion: Deeper exploration with real-world examples - Critical Evaluation: Discuss strengths, weaknesses, and gaps	5-6
Conclusion	- Summary of key points - Answer to core question (if applicable) - Suggestions for future research or implications	1-2
Q&A (Optional)	- Time for audience questions and thoughtful responses	1
References (if applicable)	- List sources used in presentation, following proper citation format (e.g., APA, MLA)	-
Closing Slide	- Thank the audience - Optionally, include contact information	-

Mid-Term Exam (2 Hours - 30 Marks)	Final Exam (3 Hours - 40 Marks)
Section A (10 Marks – 20 MCQs)	Section A (10 Marks – 20 MCQs)
Section B (10 Marks – 04 Short Questions)	Section B (10 Marks – 04 Short Questions)
Section C (10 Marks – 02 Long Questions or 01 Case Study)	Section C (10 Marks – 02 Long Questions)
	Section D (10 Marks – 01 Case Study)